Reward engineering, simulator fidelity and lap-time benchmarks

RL driving agent in OCD: Barcelona and Shanghai

Ibrahim Khaled

2 November 2026

Reward: from line tracking to time-optimal

Old (`ref_tracking`), per 50 Hz tick, e = offset to the reference line ξ_{ref}:

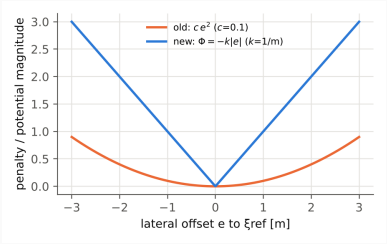
$$r = a\,\Delta s - b\,\Delta t\,\big(1 + \underbrace{\dot{u}^{\top} W \dot{u}}_{\text{jitter}}\big) - \underbrace{c\,e^{2}}_{\text{tracking}}, \qquad P_{\text{fail}} = 50 + 30\,c\,e_{\text{exit}}^{2}$$

New (`time_pbrs`, R1):

$$r = a\,\Delta s - b\,\Delta t + \big[\gamma\,\Phi(s') - \Phi(s)\big], \qquad \Phi(s) = -k\,|e(s)|,\ k = 1\,\text{m}^{-1}, \qquad P_{\text{fail}} = 100$$

1. **Jitter term removed** (logged only): top racing agents (Linesight, GT Sophy) use none.
2. **Linear $|e|$, not e^2**: constant gradient instead of flat-then-explosive.
3. **Potential-based shaping** (Ng 1999): cannot change the optimal driver; ξ_{ref} only guides early learning.

Old problem: tracking cost (175.9) exceeded lap time (123.3); the driver stayed glued to a *safe* line.



Effect of the new reward (Shanghai)

Run	Car / reward	Best clean lap 2	Own QSS (safe line)
full1	AV-21, old reward	122.85 s	133.9 s
r1_warm	AV-21, new reward, warm start	123.04 s	133.9 s
sf70h_mac_r1	sf70h_tum, new reward, from scratch	**103.24 s**	104.78 s
p1_TUM	sf70h_tum + TUM-like setup, new reward	**101.27 s**	103.56 s

- The warm start kept the old line-following habits and gained nothing.
- From scratch, the free driver **beats the QSS lap on its own safe line** by 1.5–2.3 s, because it is no longer tied to that line.
- Cost: a lower clean rate (48/194 clean deterministic evaluations in sf70h_mac_r1).

Vehicle dynamics: three cars, built in layers

	`av21` (shipped OCD)	`sf70h_tum` (27 Oct)	`sf70h_f1` (1 Nov)
Idea	Dallara AV-21 (Indy Autonomous Challenge)	AV-21 model with every value TUM's `F1_Shanghai.ini` gives	`sf70h_tum` + F1 values from papers and the FIA rules
Mass, geometry, aero	AV-21 (800 kg, 3.2 m)	TUM (733 kg, 3.6 m)	TUM
Tyre peak μ	AV-21 (1.46 / 1.16)	TUM (1.65–2.15)	TUM
Tyre slip stiffness, shape	AV-21	AV-21	PL14 (Limebeer)
Inertias	AV-21	scaled ($I \propto mL^2$)	scaled
Steering lock	0.6 rad	0.6 rad	0.30 rad
Heave springs	off	off	1.0 / 0.25 MN/m
Brakes	12 kN m, 0.50	12–15 kN m, 0.50	15 kN m, 0.58
Footprint	4.2×2.0 m	4.6×2.0 m	5.006×2.07 m

Each car is opt-in; `av21` and `sf70h_tum` stay bit-identical (85 unit tests, Shanghai `lap.csv` sha256 unchanged).

`sf70h_tum`: an F1-like car from TUM's open data

	AV-21	`sf70h_tum`
Mass	800 kg	733 kg
Wheelbase	3.2 m	3.6 m
Tyre grip (peak μ)	1.2–1.5	1.65–2.15
Engine	TUM F1	TUM F1
Aero (downforce, drag)	TUM F1	TUM F1

- Source: TUM's open F1 car file (`F1_Shanghai.ini`, laptime-simulation). Every value it provides was copied; the rest stayed AV-21.
- Biggest change: **tyre grip at F1-slick level** (about 50 % more than the AV-21 on average).
- **ERS and DRS were added to the model but are OFF in every RL run** and in our comparisons. Only TUM's 91.24 s reference uses them.
- Brakes raised from 12 to 15 kN m so the F1 grip can be used under braking.

TUM's optimal lap, driven in OCD by pure pursuit

Test: take TUM's own Shanghai lap (its racing line and speed profile, no ERS/DRS: **94.89 s**) and drive it in OCD with a simple *pure-pursuit* driver. It steers toward a point on TUM's line a short distance ahead, and its pedals follow TUM's target speed.

Car in OCD (ERS off)	Lap 2	Time on track
`sf70h_tum`, default setup, 12 kN m brakes	105.72 s	72.6 %
`sf70h_tum`, TUM-like setup (p1_TUM), 15 kN m	104.03 s	75.9 %
TUM's own lap (reference)	94.89 s	–
Our RL driver, p1_TUM (for comparison)	**101.27 s**	100 %

- The OCD car can't reproduce TUM's ideal lap: about **9–11 s slower**, and off track for a quarter of the lap. TUM's line is too tight for the car's 4.6×2.0 m outline on 30 % of its length.
- Our RL driver is **2.8 s faster** than following TUM's line, *and* stays on track the whole lap.

sf70h_f1: the next car (1 Nov)

- **Why:** further driving tests showed sf70h_tum losing most of its time on corner exits and under braking. The cause was tyre behaviour still taken from the AV-21.
- **What changed** (values from published F1 papers and the FIA 2017 rules):
  - tyre behaviour from Perantoni & Limebeer (2014), same grip level
  - F1 steering lock (17°), F1 heave springs, F1 brake balance
  - F1 car outline for the track-limit check (5.0×2.07 m)
- **Status:** tested with physics tools only. **No RL run on sf70h_f1 yet**; all RL results in this deck use sf70h_tum.
- **Open issue:** the car bounces at high speed (porpoising) because no heave-damping value is available. A value is needed before RL training.

Limebeer's optimal-control laps, and ours

Limebeer's findings (Barcelona)

- Perantoni & Limebeer 2014: **82.43 s** (82.57 s with a finer grid). Optimising the car's setup gained ≈1.5 s.
- Masouleh & Limebeer 2016, with suspension: **84.55 s**, 82.75 s after setup optimisation.
- No Limebeer lap exists for Shanghai.

What each car is

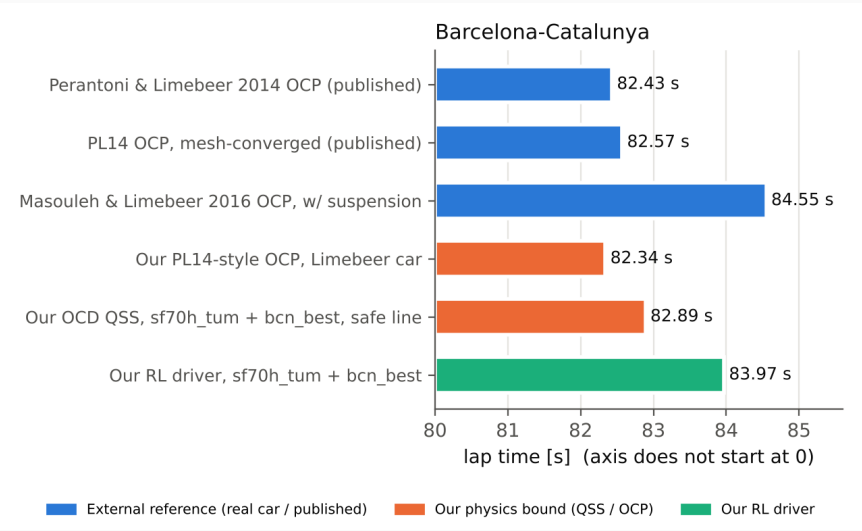
- *Limebeer car*: a 2013-era F1 car (660 kg, 560 kW).
- *Our car* (`sf70h_tum`): F1-like car from TUM's 2017 data (733 kg, 575 kW, more downforce).

Our results

Barcelona	Lap
Our OCP, Limebeer's car	82.34 s
Our OCP, our car	78.76 s
Our QSS, our car, safe line	82.89 s
Our RL driver, our car	**83.97 s**
Shanghai	
Our OCP, our car	94.25 s
TUM's simulator, same car	94.89 s
Our RL driver, our car	**101.27 s**

Our OCP reproduces Limebeer's Barcelona lap to 0.3 %, so its other results can be trusted.

Barcelona: Limebeer's OCP vs. our RL driver



Shanghai: real world vs. QSS vs. our RL driver

